Real, complex model:

$\boldsymbol{s}_{t+1} = f(\boldsymbol{s}_t, \boldsymbol{a}_t, \boldsymbol{d}_t, \boldsymbol{\epsilon}_t)$

(8 states)

Inadequate information

Practical, simplified model:

$\boldsymbol{s}'_{t+1} = f'_{\boldsymbol{\theta}_{\mathrm{mi}}}(\boldsymbol{s}'_t, \boldsymbol{a}_t, \boldsymbol{d}_t)$

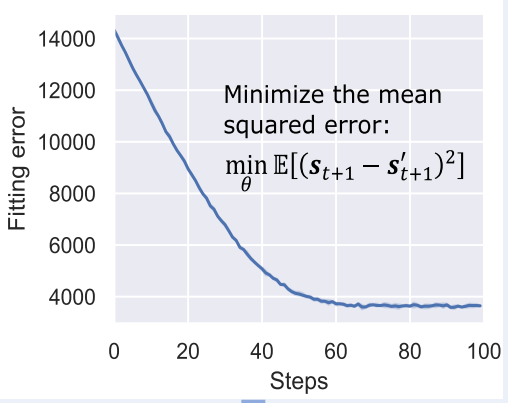
(4 states)

Model fitting

$\boldsymbol{\theta}_{\mathrm{mi}} \rightarrow \boldsymbol{\theta}_{\mathrm{mf}}$

Fitted model:

$\boldsymbol{s}'_{t+1} = f'_{\boldsymbol{\theta}_{\mathrm{mf}}}(\boldsymbol{s}'_t, \boldsymbol{a}_t, \boldsymbol{d}_t)$



Fully parameterized MPC:

$$\begin{aligned} \min \quad & \text{Prameterized costs } (\boldsymbol{\theta}_{\mathrm{ct}}) \\ \text{s.t.} \quad & \boldsymbol{s}'_{t+1} = f'_{\boldsymbol{\theta}_{\mathrm{mf}}}(\boldsymbol{s}'_t, \boldsymbol{a}_t, \boldsymbol{d}_t) \\ & \text{paramerized constraints } (\boldsymbol{\theta}_{\mathrm{cs}}) \\ \boldsymbol{\theta}_0 = & \{\boldsymbol{\theta}_{\mathrm{mf}}, \boldsymbol{\theta}_{\mathrm{ct}}, \boldsymbol{\theta}_{\mathrm{cs}}\} \end{aligned}$$

MPC-RL

$\boldsymbol{\theta}_0 \rightarrow \boldsymbol{\theta}^*$

Learned optimal parameters :

$\boldsymbol{\theta}^* = \{\boldsymbol{\theta}^*_{\mathrm{m}}, \boldsymbol{\theta}^*_{\mathrm{ct}}, \boldsymbol{\theta}^*_{\mathrm{cs}}\}$

Performance:

